MedVisionQuery: An Agentic Multimodal System for Medical Visual Question Answering with Explanation

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Abstract

Medical visual question answering (VQA) over radiology images combines perception, question interpretation, answer synthesis and verification—steps that fail in different ways and are hidden inside a single model call. We present MedVisionQuery, an agentic system in which a Haystack Agent controller decides, case by case, how to look at the image, answer, verify the answer *against the image*, and whether to re-examine a specific region, revise, abstain or finish. Perception is question-aware, the answerer and the verifier both see the image, and a different, larger model verifies. We evaluate on 200 randomly sampled VQA-RAD test questions against a single-prompt baseline, a hand-designed workflow and a fixed-pipeline ablation, all on the same answering model, scoring with a strict rule-based scorer and a blinded LLM judge. The agentic loop matches the single prompt in accuracy (64.5% vs. 63.5% strict; 72.0% vs. 70.0% judged; McNemar $p = 0.86$), clearly beats the fixed pipeline (+7.5 points, $p = 0.02$) and the workflow (28.0%), and is substantially more reliable: its verification and self-consistency signals predict errors, it abstains on hard cases, and only 2% of its answers change across re-runs versus 12% for the single prompt—at about 45$\times$ the cost. Code is available at `https://github.com/clarence-ck/med-vqa-rad-agentic`.

1 Problem Definition and Motivation

Medical VQA is a fundamentally multi-stage reasoning problem [1]. As established in the literature, a complete medical VQA system must address at minimum an image encoder, a question encoder, a multimodal fusion component, and an answer prediction module [1]—but these four components conceal distinct reasoning sub-problems that interact in clinically important ways. Given a radiology image and a clinical question, a system must separately solve: (1) *perception*—what is visible in the image that bears on the question; (2) *question interpretation*—what type of reasoning and answer is required, since different question types demand fundamentally different skills [2, 3]; (3) *synthesis*—how to integrate visual evidence into a grounded answer; (4) *verification*—whether the answer is actually supported by the visual evidence, a stage explicitly modelled in recent work as a "Confirm" module for cross-modal validation [5]; and (5) *control*—deciding, from what verification finds, whether to re-examine the image, revise the answer, abstain or stop.

These sub-tasks interact in non-trivial ways that make single-prompt approaches insufficient: what to look at depends on the question; the verifier must check cross-modal consistency between the image and the generated answer; and clinical safety requires explicit guardrails, abstention and disclaimers when confidence is low.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

Existing work: classification approaches. Early and still-prevalent medical VQA methods frame the task as multi-class classification over a fixed answer vocabulary [1]. The VQA-Med challenge [6] established the dominant paradigm: an image encoder and a question encoder produce fused features passed to a classification head. MEVF [7] extended this with meta-learning (MAML) and a denoising autoencoder (CDAE), feeding image features into a bilinear attention network (BAN) followed by a classifier. PMC-VQA [8] proposed MedVInT-TE, similarly mapping image–question pairs to answers via an encoder and classification head. Classification approaches are computationally efficient on close-ended questions but constrained to answers seen during training and perform poorly on open-ended questions requiring free-form responses.

Existing work: generation approaches. To handle open-ended questions, a parallel line of work adopts autoregressive generation. Bazi et al. [9] proposed a full transformer encoder–decoder architecture demonstrating gains on VQA-RAD and PathVQA. PMC-VQA's MedVInT-TD [8] further demonstrated that an LLM decoder can match or exceed classification-based performance when fine-tuned on large biomedical corpora. LLaVA-Med [10], accepted at NeurIPS 2023, adapted the general-domain LLaVA architecture to biomedicine via curriculum learning on PMC-15M figure–caption pairs and GPT-4-generated instruction-following data. RadFM [11] scaled this into a generalist radiology foundation model capable of VQA, report generation, and rationale diagnosis. R-LLaVA [12] extended LLaVA-Med by injecting clinician-annotated regions of interest into the CLIP encoder, improving spatial and contextual comprehension.

Existing work: hybrid approaches. Q2ATransformer [2] proposed a unified transformer framework introducing learnable candidate answer embeddings as a decoder, querying the fused image–question representation for each answer class. Although framed as classification, candidate answer embeddings interact through transformer attention, incorporating aspects of generative reasoning. This hybrid achieves state-of-the-art performance on VQA-RAD and PathVQA.

The gap that remains. Despite this progress, *all* prior approaches—whether classification [6–8], generation [9–12], or hybrid [2]—treat the task as a single end-to-end model invocation, making it impossible to localise failures, adapt the reasoning process to what the image shows, or enforce clinical safety constraints. Even recent works that acknowledge the need for verification [5] or knowledge grounding [4] address these as training objectives within a single model, rather than as explicit, auditable steps. MedVisionQuery fills this gap with an agent controller that coordinates question-aware perception, answering and image-grounded verification, deciding per case whether to re-examine the image, revise, abstain or finish, with structured failure handling.

We target the **VQA-RAD** benchmark [13], a clinically generated radiology VQA dataset; its Hugging Face release contains 2,244 image–question pairs (1,793 train / 451 test) across close-ended and open-ended question types. We evaluate on a seeded random sample of 200 test questions; no training data is used.

2 Agent Architecture

Controller loop. A fresh Haystack `Agent` is created per request (Figure 1). Its five tools are bound to per-request working memory holding the image, the question analysis, and every look, answer and check; the controller sees only short text results and chooses the next call. Unlike our earlier fixed pipeline, the tool sequence depends on what verification finds: on the 200 evaluation cases the controller finished directly in 141 cases, re-examined a region in 19, revised its answer in 29 and abstained in 11. Budgets cap the loop at two focused re-looks, three answers and 16 steps, with a safety net that checks the latest answer if the controller stalls.

Roles. `analyze_question` (planner input) classifies the question type and expected answer format. `look(focus)` (perception) examines the image *with the question in mind*; with a focus, it re-examines a named region; the first look rejects non-radiology images. `answer(guidance)` (executor) answers from the image plus all observations, applying the verifier's guidance on revisions. `check` (evaluator) verifies the answer against the image with five checks, returns a status (verified, partially verified, low confidence, unsupported) and recommends *accept*, *relook* with a focus, *reanswer* with guidance, or *abstain*. `finish(abstain)` ends the episode and is refused while the latest answer is unchecked.

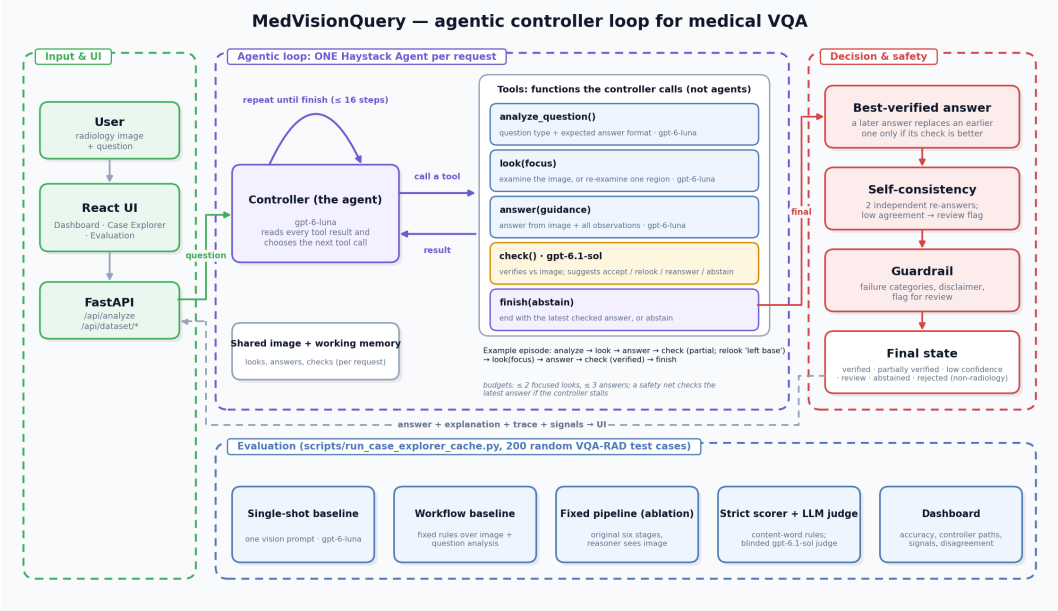


Figure 1: MedVisionQuery. One Haystack Agent (the controller) calls five tools in a loop, reading each result before choosing the next: question analysis, question-aware perception (`look`), answer synthesis, image-grounded verification (`check`) and `finish`. The tools are not agents; `check` only recommends a re-look, a revised answer, abstention or acceptance, and the controller decides. A decision-and-safety stage keeps the best-verified answer, measures self-consistency and applies guardrails.

Model and tool choices. All roles use OpenAI models, sized by need and configurable per role. The controller and question analysis make simple decisions (`gpt-6-luna`, no/low reasoning effort); perception and answering use `gpt-6-luna` with medium effort, after a pilot showed that giving the answerer the image mattered far more than model size (50%→70% on 20 cases). Verification uses a different, larger model (`gpt-6.1-sol`) so that the answering model does not verify itself. Haystack's `Agent/Tool` abstraction provides the tool-calling loop and exit conditions; keeping the image in per-request memory rather than in the controller's context keeps the controller cheap. An earlier retrieval stage over SLAKE [15] was removed: it returned other patients' question–answer pairs, which contradicted yes/no and location answers and gave no measurable benefit on perception-bound questions.

Baselines and ablation. The *single-shot baseline* sends the image and question in one prompt. The *workflow baseline* is a fixed chain—a question-agnostic image description, question analysis, then hand-written rules per question type. The *fixed-pipeline ablation* runs our original six stages (interpret, analyse, filter, route, reason, verify, with retry) in a fixed order with the image given to the reasoner, isolating what the controller's decisions add. A demo UI (Section 6) exposes all systems.

3 Multimodal Grounding

The system uses two modalities, the radiology image and the text question, and they interact at every step that determines the answer: the question (and any focus) steers what `look` examines; `answer` and `check` receive the image directly alongside the text observations; only the controller works purely in text. Our earlier design described the image once, *without* the question, and passed only that text downstream; in a 20-case pilot all four failures of that design were "cannot be determined from the image description", and all four were answered correctly once the reasoner saw the image. Because the verifier sees the image, its "explanation grounded in evidence" and "no contradictions" checks are genuine cross-modal agreement checks.

Table 1: Accuracy (%) on 200 random VQA-RAD test questions (same answering model for all systems). Open-ended accuracy is strict/judged. Time is per-case latency in seconds (mean/median), measured with four cases running concurrently; the whole 200-case run of all four systems took 40 minutes of wall-clock time. Cost is the mean API cost per case.

System	Strict	Judged	Close (111)	Open (89)	Time (s)	Cost
Single-shot baseline	63.5	70.0	76.6	47.2 / 61.8	4.4 / 3.6	\$0.0002
Workflow baseline	28.0	31.0	42.3	10.1 / 16.9	9.0 / 8.7	\$0.0005
Agentic loop	**64.5**	**72.0**	76.6	49.4 / 66.3	37.2 / 29.3	\$0.0094
Fixed pipeline (ablation)	57.0	61.5	68.5	42.7 / 52.8	38.1 / 29.4	\$0.0070

Table 2: Left: accuracy (strict, %) by the path the controller took. Right: agentic accuracy by its own signals.

Path	n	Agentic	Single	Pipeline	Signal	Agentic
direct	141	74	72	67	verified (169)	70
relook	19	47	37	37	partially verified (25)	40
reanswer	29	55	45	31	unsupported/low (6)	0
abstained	11	–	45	36	consistency high (162)	72
					consistency med/low (27)	40–53

4 Control and Failure Handling

Detection. Image-grounded verification (five checks and a status), seven failure categories (*unsupported claim*, *low confidence*, *visual contradiction*, *partially verified*, *question–answer mismatch*, *generation unstable*, *parser failure*) and an eighth, *self-inconsistent*, from resampling. **Recovery.** Instead of re-running the whole pipeline, the controller re-examines the region the verifier names or revises the answer with the verifier's guidance. A later answer replaces an earlier one only if its check is strictly better, because focused re-looks can over-read. **Self-critique.** After finishing, the answerer is re-asked independently twice; low agreement adds a failure category and flags the case for review. **Fallback.** The controller may abstain (reported as an abstention, never coerced into "no"); a guardrail adds a disclaimer and a review flag for low-confidence, unsupported, high-severity or self-inconsistent cases; non-radiology or unreadable images are rejected before any answer is produced.

5 Evaluation

Setup. 200 VQA-RAD test questions drawn at random (seed 42; 111 close-ended, 89 open-ended); all four systems run on the same cases with the same answering model. *Strict scoring*: yes/no answers must start with the correct word; binary terms must name the correct term and not its opposite; counts must match; open-ended answers must contain every content word of the reference (aliases and plurals folded; $\geq$60% for references longer than three words), with left/right conflicts and denials counted wrong and embedding similarity [14] only as a strict fallback. Because strict scoring rejects correct paraphrases, a *blinded LLM judge* (`gpt-6.1-sol`, answers shuffled and anonymised) re-assesses strict-wrong open-ended answers; a 30-verdict spot check agreed on 29/30 credit decisions. Paired comparisons use the exact McNemar test; 95% intervals are about $\pm$6.7 points. The baseline and agentic loop were re-run on 50 cases to measure stability. API cost of the evaluation: \$4.08 (\$3.43 runs, \$0.54 repeats, \$0.11 judge). Actual wall-clock time: 40 min for the main run (four cases at a time), 35 min for the 50 sequential repeats and 6 min for the judge.

Results. Table 1. The agentic loop and the single prompt are statistically indistinguishable (17 vs. 15 discordant cases, $p = 0.86$; judged $p = 0.61$). The controller's decisions clearly help relative to the same models in a fixed order (26 vs. 11, $p = 0.02$; judged 32 vs. 11, $p = 0.002$), and both far exceed the hand-designed workflow. Gains concentrate where the controller acts: on cases it re-examined it scored 47% versus 37% for the single prompt, and on revised cases 55% versus 45% (Table 2).

Failure case analysis. *Self-consistency:* two independent re-answers per case; high consistency means 72% accuracy, medium/low 40–53%, so low consistency marks uncertainty rather than certain error. *Cross-modal agreement:* because the verifier sees the image, verification status is predictive: verified answers are 70% correct, unsupported or low-confidence answers 0%. *Verifier model:* a larger model verifying a smaller one steers 59 of 200 episodes away from the direct path. *Disagreement:* the three systems disagree on 65% of cases; when exactly one system is right, it is the agentic loop 16 times, the single prompt 12 and the workflow 7. *Stability:* on 50 repeated cases, 12% of single-prompt answers flipped between right and wrong, versus 2% for the agentic loop. Four patterns dominate the errors: (i) a shared perception blind spot, where looker and verifier misread the image the same way (e.g. bowel gas read as free air after two focused re-looks); (ii) over-reading on re-looks, limited by keeping the best-verified answer; (iii) benchmark items whose question does not match the image, which every system fails; and (iv) workflow rigidity, where fixed rules answer "Brain" or "X-ray" when a specific finding is needed.

6 User Interface

A React frontend exposes three views: **Dashboard** (upload an image, compare the agentic loop with both baselines, inspect the trace, verification checks and failure categories), **Case Explorer** (browse the 200 evaluation cases with full agent traces) and **Evaluation** (strict and judged accuracy for all four systems, accuracy by question type and by controller path, disagreement and a per-case table).

7 Reflection

When agentic behaviour is beneficial. Not for raw accuracy on this benchmark: a strong single prompt on the same model is as accurate. The agentic system pays off when answers must be *trustworthy*: its verification and consistency signals identify likely errors, it abstains on cases where the single prompt is wrong more often than right, it recovers on hard cases by re-examining specific regions, its answers are far more stable across runs, and every decision is auditable. Against a fixed multi-stage pipeline with identical models, letting a controller decide is clearly better—the benefit comes from *adaptive control*, not from decomposition alone.

Limitations. About $45\times$ the cost and $8\times$ the latency of a single prompt for equal accuracy. The verifier shares the perception weaknesses of vision models, so agreement is not correctness. Two hundred cases give ± 6.7-point intervals and noisy per-type estimates; the LLM judge is an upper bound, spot-checked but not human-annotated. A retrieval stage did not help perception-bound questions and was removed; retrieval may still help questions that need knowledge not visible in the image, which VQA-RAD rarely contains.

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